FieryPath Developer Guide

Fiery DFE-Press Interface Developer Guide





Revision History

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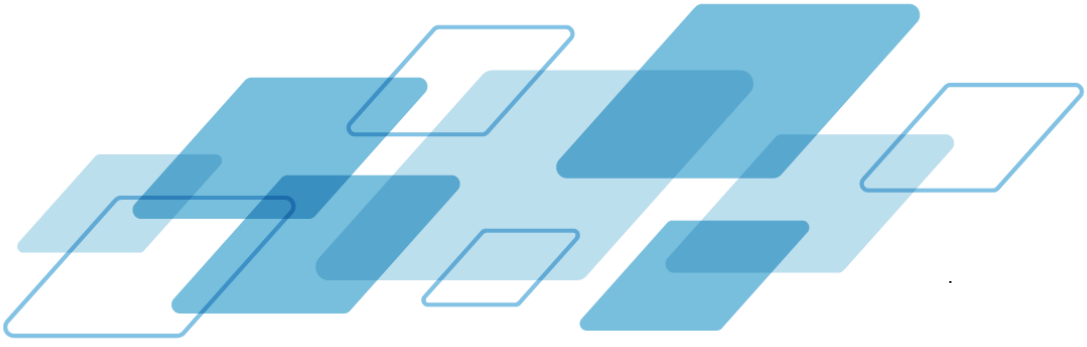
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1 Overview

This document is intended to serve as a reference for developers wanting to implement a Printer Controller that communicates with **Fiery DFE** (Digital Front End) using **FieryPath**. As described in the specification, communication between the DFE (i.e. server) and Printer Controller (i.e. client) are split into two categories:

- **Control Path**: the interface between Fiery DFE (Digital Front End) and the part of your product's internal printer software responsible for handling the real-time messages/events generated by the DFE that are required to print a job and the individual images within that job.
- **Data Path**: the interface for handling the transfer of raster image data from the DFE to your printer.

This document aims to illustrate the things your Printer Controller's **Control Path** should do based on the real-time messages/events generated by Fiery DFE and, if your product implements one, your **Printer UI** (the user-facing software used by a Printer Operator and responsible for controlling the Printer – i.e. starting/stopping the printer, uniformity/nozzle-out corrections, etc.). In addition, the document aims to provide information regarding what a Printer Controller: 1) **must** implement, 2) **might** implement (and why), and 3) how it **should** be implemented (or not implemented) to avoid common pitfalls.

A critical point of emphasis: while your product may develop several components to integrate with **Fiery DFE**, including a **Control Path** and a **Job UI** (the user-facing software responsible for controlling job properties, ordering, submission, cancellation, etc.), there is a clear, logical separation in functionality of these components, and therefore, there should be a corresponding clear, logical separation in their design, architecture, and implementation. In addition, given the clear, logical separation in purpose of the **Fiery API** (i.e. the API responsible for controlling job properties, ordering, submission, cancellation, etc.) and the **FieryPath** (i.e. the API responsible for handling the real-time messages/events generated by the DFE that are required to print a job and the individual images within that job), your product's **Job UI** software should not use the **FieryPath API**, and similarly, your Printer Controller's **Control Path** should not use the **Fiery API**. These two pieces of software, while both extremely critical for your product, are not linked in any way, and should be designed as such.

2 Control Path

The **Control Path** is broken-up into several sections: Design, Actions, and Printer UI.

2.1 Design

The details of your Printer Controller's **Control Path** design (i.e. language, structure, etc.) is obviously up to you, but it **should** implement the concept of a state machine and it **should** support most (if not all) of the states listed below:

- Initialize: used to indicate that your Printer Controller is in the process of "initializing".
- Wait: used to indicate that your Printer Controller has "initialized" but is waiting for the Printer Operator to issue a **Start** action from your **Printer UI** (provided your product has implemented this UI).
- Ready: used to indicate that your Printer Controller is "ready" to process/print jobs.
- Print: used to indicate that your Printer Controller is in the process of "printing" jobs.
- Suspend: used to indicate that your Printer Controller has "suspended" the process of "printing" jobs.
- Recover: used to indicate that your Printer Controller is in the process of "recovering" a job, so that any images that failed to completely print when a "suspend" was issued will be guaranteed to be re-printed upon issuing a "resume".
- Cancel: used to indicate that your Printer Controller is in the process of "canceling" a currently "printing" job.

The figure below shows an example of a Printer Controller's **Control Path** state diagram, along with **FieryPath** events and expected Control Path actions.

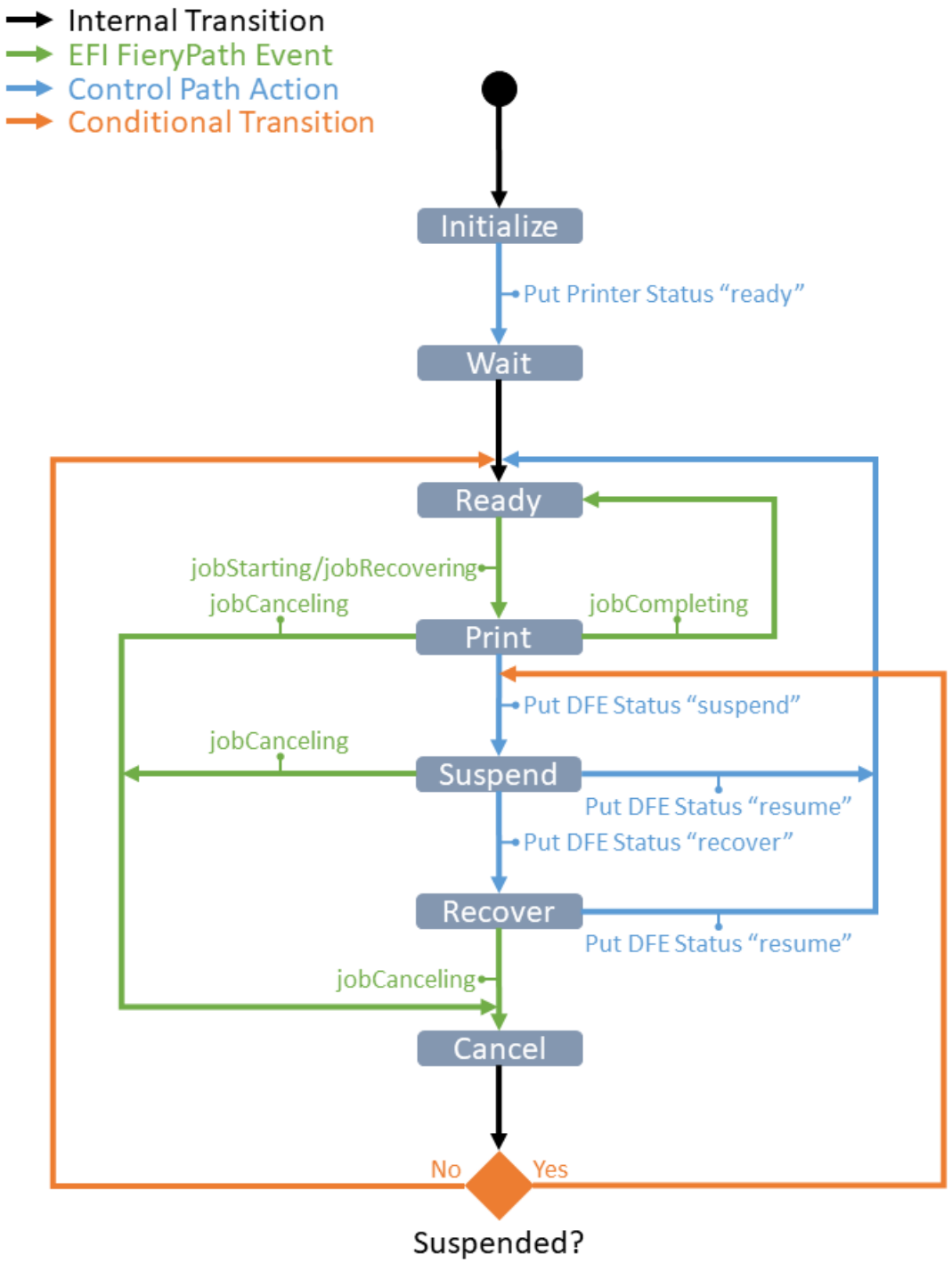


Figure 1: *Control Path state diagram and FieryPath events.*

2.1.1 Initialize

Upon start-up, your Printer Controller's **Control Path** transitions to the **Initialize** state and does whatever is necessary to initialize itself. This includes executing the **Start-up** sequence (see the **FieryPath** interface specification) and concludes upon performing the **Set Printer Status** action with the "state" element set to "ready" and transitioning to the **Wait** state. Please see the Initialization sub-section under the Actions section later in this document for more details.

2.1.2 Wait

After completing the necessary actions in the **Initialize** state, your Printer Controller's **Control Path** transitions to the **Wait** state. At this point, your Printer Controller is ready to print any incoming job, but you are awaiting the Printer Operator to indicate their readiness by issuing the **Start** action (provided your product has implemented a **Printer UI**). Once the Printer Operator issues a **Start** action, your Printer Controller's **Control Path** transitions to the **Ready** state.

2.1.3 Ready

Upon entering the **Ready** state, your Printer Controller's **Control Path** is ready to handle **jobStarting** or **jobRecovering** events. When your Printer Controller's **Control Path** receives either event, your Printer Controller does everything necessary to begin printing the given job and transitions to the **Print** state. Please see the Printing sub-section under the Actions section later in this document for more details.

2.1.4 Print

Upon entering the **Print** state, your Printer Controller's **Control Path** is ready to handle **imageStarting**, **imageCompleting**, and **jobCompleting** events (and even other **jobStarting** or **jobRecovering** events from other jobs) to successfully print the given job(s). Please see the Printing sub-section under the Actions section later in this document for more details.

2.1.5 Suspend

When "suspended", Fiery DFE will stop issuing **imageStarting** events, but it will continue issuing **imageCompleting** and **jobCompleting** events. This is where your Printer Controller does whatever is necessary to prepare for printing again. From the **Suspend** state, your Printer Controller's **Control Path** may transition to the **Ready** state, the **Recover** state, or the **Cancel** state (depending on the design of your Printer Controller and what Printer Operator UI actions you allow). Please see the Suspension/Recovery sub-section under the Actions section later in this document for more details.

2.1.6 Recover

Upon entering the **Recover** state, your Printer Controller does whatever is necessary to be ready to recover printing the currently "suspended" job(s). Please see the Suspension/Recovery sub-section under the Actions section later in this document for more details.

2.1.7 Cancel

Upon entering the **Cancel** state, your Printer Controller does whatever is necessary to cancel the given job. From the **Cancel** state, your Printer Controller's **Control Path** may transition to either the **Suspend** state or **Ready** state, depending on whether your Printer Controller's **Control Path** was previously in **Suspend** state or not. Please see the Cancelation sub-section under the Actions section later in this document for more details.

2.2 Actions

The **Actions** section is broken-up into several sub-sections: Initialization, Printing, Suspension/Recovery, Cancelation, and Periodic/As-Needed.

2.2.1 Initialization

The **Initialization** section covers the actions/commands that are performed by your Printer Controller's **Control Path** once, while in the **Initialize** state. They include:

- Log-in
- Configuration
- Status

2.2.1.1 Log-in

The **Log-in** action (execute a POST at https://<fiery_ip>/fierypath/v1/session/login) **must** be implemented by all Printer Controllers. This is the first **FieryPath** action your Printer Controller's **Control Path** should execute during initialization/start-up. To successfully perform the **Log-in** action, each product must pass in an appropriate "username", "password", and "access token" (access tokens may be acquired by contacting your **Fiery** Product Team). If the DFE is in the process of starting-up/initializing, the **Log-in** action may not initially succeed, so your Printer Controller **must** be prepared to retry the **Log-in** action until it does. Upon a successful **Log-in** action, the DFE will return a "session token" that will be used in all subsequent **FieryPath** actions. As a required step to logging-in, your Printer Controller **must** establish a WebSocket connection with the DFE – allowing it to receive **FieryPath** events. As with **Log-in**, the Printer Controller must be prepared to retry the WebSocket connection until it succeeds. The Printer Controller **should** also set its status to "initializing" at this point. Although this is an optional step, setting it at this time will allow a graceful recovery in the event the Printer Controller has abruptly restarted.

2.2.1.2 Configuration

Once your Printer Controller's **Control Path** has successfully executed the **Log-in** action, there are several **Configuration** actions that your Printer Controller **might** implement during **Initialization/Start-up**, but none of them are required to print. If your product supports a particular feature/option, then your Printer Controller **must** implement the corresponding action for the feature function correctly. The feature/option and their action(s) are described below:

2.2.1.2.1 OEM Configuration

If your product supports product-specific configuration information (or if needed dynamic-ink support), then your Printer Controller **must** transfer it. This will allow the DFE to properly configure/support various product-specific behaviors. Since the content of this information is specific to your printer, please check with your **Fiery** Product Team to know the structure/content of the data to be sent. To properly transfer product-specific configuration information, your Printer Controller **should** do the following:

1. Transfer new product-specific configuration (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/oem/configuration).

2.2.1.2.2 Installed Inks

If your product supports communicating information about its installed inks (in particular, sending current ink level information), then your Printer Controller **must** enumerate them. This will allow the DFE to maintain ink level information that can be displayed by Command WorkStation (CWS) and other UIs. To properly enumerate the installed inks, your Printer Controller **should** do the following:

1. Attempt to get the list of "currently installed" inks (execute a GET at https://<fiery_ip>/fierypath/v1/system/printer/inks). If successful:
    a. Enumerate through the "currently installed" inks and delete them (execute a DELETE at https://<fiery_ip>/fierypath/v1/system/printer/inks/<id> for each ink to be deleted).
2. Another option is to delete all the installed inks without enumeration (execute a DELETE at https://<fiery_ip>/fierypath/v1/system/printer/inks to delete all inks)
3. Enumerate through your existing inks and add them (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/inks for each ink to be added).
4. Set initial status for each installed ink (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/inks/<id>/status for each ink).

2.2.1.2.3 Installed Inputs

If your product supports communicating information about its installed inputs (in particular, sending current input level information), then your Printer Controller **must** enumerate them. This will allow the DFE to maintain input level information that can be displayed by Command WorkStation (CWS) and other UIs. To properly enumerate the installed inputs, your Printer Controller **should** do the following:

1. Get the list of "currently installed" inputs (execute a GET at https://<fiery_ip>/fierypath/v1/system/printer/inputs). If successful:
    a. Enumerate through the "currently installed" inputs and delete them (execute a DELETE at https://<fiery_ip>/fierypath/v1/system/printer/inputs/<id> for each input to be deleted).
2. Another option is to delete all the installed inputs without enumeration (execute a DELETE at https://<fiery_ip>/fierypath/v1/system/printer/inputs to delete all inputs)
3. Enumerate through your existing inputs and add them (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/inputs for each input to be added).
4. Set initial status for each installed input (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/inputs/<id>/status for each input). Not all input attributes are required (because they are product-specific), so please check with your **Fiery** Product Team to know which attributes should be sent.

2.2.1.2.4 Installed Finishers

If your product supports optional installed finishers, then your Printer Controller **must** enumerate it. This will allow the DFE to properly enable/disable job property features associated the optional installed finishers. To properly enumerate the installed finishers, your Printer Controller **should** do the following:

1. Get the list of "currently installed" finishers (execute a GET at https://<fiery_ip>/fierypath/v1/system/printer/finishers). If successful:
    a. Enumerate through the "currently installed" finishers and delete them (execute a DELETE at https://<fiery_ip>/fierypath/v1/system/printer/finishers/<id> for each finisher to be deleted).
2. Another option is to delete all the installed finishers without enumeration (execute a DELETE at

https://<fiery_ip>/fierypath/v1/system/printer/finishers to delete all finishers)

3. Enumerate through your existing finishers and add them (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/finishers for each finisher to be added).

2.2.1.2.5 Installed Outputs

If your product supports optional installed outputs, then your Printer Controller **must** enumerate it. This will allow the DFE to properly enable/disable job property features associated the optional installed outputs. To properly enumerate the installed outputs, your Printer Controller **should** do the following:

1. Get the list of "currently installed" outputs (execute a GET at https://<fiery_ip>/fierypath/v1/system/printer/outputs). If successful:

    a. Enumerate through the "currently installed" outputs and delete them (execute a DELETE at https://<fiery_ip>/fierypath/v1/system/printer/outputs/<id> for each output to be deleted).

2. Another option is to delete all the installed outputs without enumeration (execute a DELETE at https://<fiery_ip>/fierypath/v1/system/printer/outputs to delete all outputs)

3. Enumerate through your existing outputs and add them (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/outputs for each output to be added).

2.2.1.2.6 Localization

If your product supports localization strings, then your Printer Controller **must** transfer it. This will allow the DFE to properly display product-specific, localized strings whenever your Printer Controller's **Control Path** updates the "printer status". To properly transfer localized strings, your Printer Controller **should** do the following:

1. Check to see if "current localization" exists (execute a GET at https://<fiery_ip>/fierypath/v1/system/printer/localization). If successful:

    a. Delete "current localization" (execute a DELETE at https://<fiery_ip>/fierypath/v1/system/printer/localization).

2. Transfer new localization (execute a POST at either https://<fiery_ip>/fierypath/v1/system/printer/localization or https://<fiery_ip>/fierypath/v1/system/printer/localization/upload).

2.2.1.2.7 OEM Corrections

If your product supports uniformity and/or nozzle-out corrections, then your Printer Controller **must** transfer it (provided your Printer Controller has a current/up-to-date set of uniformity and/or nozzle-out corrections). This will allow the DFE to properly compensate the image data according to your printer's uniformity and/or nozzle-out correction values. Given that this data can be large and difficult/impossible to persist on your printer, your Printer Controller **should** always send its uniformity and/or nozzle-out corrections at **Initialization/Start-up**. The DFE will determine if the corrections differ from what's currently installed (if any) and update accordingly. To properly transfer uniformity and/or nozzle-out corrections, your Printer Controller **should** do the following:

1. Upload the necessary nozzle-out correction files (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/nozzleOut for each set of printheads).

2. Apply uploaded nozzle-out correction files (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/nozzleOut).

3. Upload the necessary uniformity correction files (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/uniformity for each set of printheads).

4. Apply uploaded uniformity correction files (execute a PUT at

https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/uniformity).

2.2.1.3 Status

Once your Printer Controller's **Control Path** has successfully executed any of the previous **Configuration** action(s), there is a **Status** action that your Printer Controller **must** implement during **Initialization/Start-up** (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/status with "state" set to "ready"). The DFE will NOT send any Printing events until your Printer Controller's **Control Path** is "ready".

2.2.2 Printing

Once your Printer Controller's **Control Path** has successfully set its status to "ready", it will be in the **Ready** state and able to receive "printing" events. The **Printing** section covers the sequential, but asynchronous **events** that are handled by your Printer Controller's **Control Path** every time a job is printed. They include:

- jobStarting
- jobInterrupting
- jobRestarting
- jobRecovering
- imageStarting
- imageCompleting
- imageCanceling
- jobCompleting
- jobCanceling

2.2.2.1 jobStarting

The **jobStarting** event is used to inform your Printer Controller of an impending print job. The event contains details about the job that will allow your Printer Controller's **Control Path** to check and confirm that your printer is properly configured to print the impending job. Once confirmed, your Printer Controller **must** respond (execute a PUT at https://<fiery_ip>/fierypath/v1/jobs/<id>/status) with a "status" JSON object containing "state" set to "start" in order for the print job to proceed.

These events can be used to create "job" objects in your Printer Controller's **Control Path**, that will allow you to manage the images associated with a job.

2.2.2.2 jobInterrupting

The **jobInterrupting** event is used to inform your Printer Controller that a "rush" job is ready to print and your Printer Controller will stop receiving new imageStarting events for the current job until all "rush" jobs are processed. The event contains the same details as the **jobCompleting** event. No response is required as this is a purely informational event.

2.2.2.3 jobRestarting

The **jobRestarting** event is used to inform your Printer Controller that a previously "interrupted" job (to allow one or more "rush" jobs to print) is ready to restart. The event contains the same details as the **jobStarting** event. No response is required as this is a purely informational event.

2.2.2.4 jobRecovering

The **jobRecovering** event is used to inform your Printer Controller that a previously "suspended" job has been "recovered" and is ready to print. The event contains the same details as the **jobStarting** event and should be handled as described above in the jobStarting sub-section.

2.2.2.5 imageStarting

The **imageStarting** event is used to inform your Printer Controller of an impending image for the given print job. Your Printer Controller **must** respond (execute a PUT at https://<fiery_ip>/fierypath/v1/jobs/<id>/status) with a "status" JSON object containing "state" set to "start" and "id" set to the image ID received in the corresponding **imageStarting** event in order for the image to be printed.

These events can be used to create "image" objects in your Printer Controller's **Control Path** and associate them with the appropriate "job" object. The event contains details about the image that your Printer Controller's **Control Path** may want to track as it passes through your printer. Remember, however, these events are purely informational and **must not** be used to throttle the print job (i.e. control the speed/timing) – so responding to each **imageStarting** event as quickly as possible is **critical**.

Since the **events** defined by the **FieryPath** specification are asynchronous, it is possible that your Printer Controller's **Control Path** will receive one or more **imageStarting** events prior to handling the **jobStarting** event. This is expected – we don't want to unnecessarily serialize events because we need to ensure both your printer and the DFE are running as fast as possible in order to print jobs at the high speeds your printer supports.

If this image is part of a group, you need to ensure that all images which belong to the same group are "started" in the same order as they are received. Additionally, if an offset if specified, it should be the same for all images belonging to the same group.

2.2.2.6 imageCompleting

The **imageCompleting** event is used to inform your Printer Controller that the given image has been transferred to your printer by the **Data Path**. The event contains details about the image that your Printer Controller's **Control Path** may want to track as it passes through your printer. Once your printer confirms the image has been successfully printed, your Printer Controller **must** respond (execute a PUT at https://<fiery_ip>/fierypath/v1/jobs/<id>/status) with a "status" JSON object containing "state" set to "complete" and "id" set to the image ID received in the corresponding **imageCompleting** event in order for the image to be marked as "printed" in the DFE.

These events can be used to delete the associated "image" object in your Printer Controller's **Control Path**, because this image will not be referenced again.

2.2.2.7 imageCanceling

The **imageCanceling** event is used to inform your Printer Controller that the given image has canceled and will **not** be transferred to your printer by the **Data Path**. No response from your Printer Controller's **Control Path** is necessary, but if your software created an associated "image" object, it can now be removed because this image will not be referenced again.

2.2.2.8 jobCompleting

The **jobCompleting** event is used to inform your Printer Controller that the given job is coming to an end and all **imageStarting** events associated with the given job have been sent. Most likely, your Printer Controller's **Control Path** will still be receiving **imageCompleting** events and your Printer Controller **must** continue to handle them as expected, once your printer confirms all images for the given job have been successfully printed, your Printer Controller **must** respond (execute a PUT at https://<fiery_ip>/fierypath/v1/jobs/<id>/status) with a "status" JSON object containing "state" set to "complete" in order for the job to be marked as "printed" in the DFE.

These events can be used to delete the associated "job" object (once all associated "image" objects have been deleted due to handling **imageCompleting** and **imageCanceling** events) in your Printer Controller's **Control Path**, because this job will not be referenced again.

2.2.2.9 jobCanceling

The **jobCanceling** event is used to inform your Printer Controller that the given job has been canceled and no more **imageStarting** events associated with the job will be sent. Since the **events** defined by the **FieryPath** specification are

asynchronous, it is possible that your Printer Controller **might** receive one or more **imageCompleting** events for the given "canceled" job, but this is expected. At this point, it is up to your Printer Controller to decide which (if any) of the images associated with outstanding **imageCompleting** events should be allowed to finish printing. Once your printer confirms all expected images for the "canceled" job have been successfully printed, your Printer Controller **must** respond (execute a PUT at https://<fiery_ip>/fierypath/v1/jobs/<id>/status) with a "status" JSON object containing "state" set to "cancel" in order for the job to be marked as "cancel" in the DFE.

These events can be used to delete the associated "job" object (once all associated "image" objects have been deleted due to handling **imageCompleting** and **imageCanceling** events) in your Printer Controller's **Control Path**, because this job will not be referenced again.

2.2.3 Suspension/Recovery

Once "suspended", the Fiery DFE will stop issuing **imageStarting** events, but it will continue issuing **imageCompleting** events (but only for the **imageStarting** events your Printer Controller's **Control Path** has properly responded to) and **jobCompleting** events (if appropriate). This is where your Printer Controller does whatever is necessary to prepare for printing again. Depending on what Printer Operator UI actions are allowed, your Printer Controller's **Control Path** may have to support **Suspend (Immediate)**, **Suspend (Normal)**, or both.

If supporting **Suspend (Immediate)**, your Printer Controller's **Control Path** will stop the printer immediately, ignoring any subsequent **imageCompleting** and **jobCompleting** events. Because your printer will **NOT** successfully print all outstanding images, then your Printer Controller **must** perform a "recovery" in order to pick-up printing where things left off. This requires performing the **Set DFE Status** action (see the **FieryPath** interface specification) with the "action" element set to "recover" and transitioning to the **Recover** state.

If supporting **Suspend (Normal)**, your Printer Controller's **Control Path** will wind-down the printer while attempting to successfully print all outstanding images. This means your Printer Controller **must** properly respond to all **imageCompleting** and **jobCompleting** events, as needed. If your Printer Controller successfully prints all outstanding images, then your Printer Controller's **Control Path** will be able to continue printing from where things left off by performing the **Set DFE Status** action (see the **FieryPath** interface specification) with the "action" element set to "resume" and transitioning to the **Ready** state. If, however, your Printer Controller was unable to successfully print all outstanding images, then your Printer Controller **must** perform a "recovery", as described above for **Suspend (Immediate)**, to pick-up printing where things left off.

In addition, your Printer Controller's **Control Path** needs to be ready to handle several asynchronous events and actions from the **Suspend** state. This includes the **jobCanceling** event and the Printer Operator's **Abort** or **Start** actions. Upon receiving the Printer Operator's **Start** action, your Printer Controller's **Control Path** performs the **Set DFE Status** action (see the **FieryPath** interface specification) with the "action" element set to "resume" and transitions to the **Ready** state. Upon receiving the Printer Operator's **Abort** action, your Printer Controller's **Control Path** performs the **Set DFE Status** action (see the **FieryPath** interface specification) with the "action" element set to "abort" or "abortall" and transitions to the **Cancel** state. If your Printer Controller's **Control Path** receives the **jobCanceling** event (the given job was canceled through some other mechanism), your Printer Controller's **Control Path** simply transitions to the **Cancel** state.

2.2.4 Cancelation

Upon entering the **Cancel** state, your Printer Controller does whatever is necessary to cancel the given job. Once complete, your Printer Controller's **Control Path** will transition to another state. If **NOT** previously in the **Suspend** state, your Printer Controller's **Control Path** will transition directly to the **Ready** state. If previously in the **Suspend** state, however, your Printer Controller's **Control Path** will first transition back to the **Suspend** state. Once there, your Printer Controller's **Control Path** needs to be ready to handle the Printer Operator's **Start** action, as described above in the Suspension/Recovery sub-section.

2.2.5 Periodic/As-Needed

The **Periodic/As-Needed** section covers the actions/commands that are performed by your Printer Controller's **Control Path** on a periodic, as-needed basis. They include:

- Product-Specific Configuration
- Ink Status
- Input Status
- Uniformity/Nozzle-Out Corrections
- Calibration/Profiling Corrections

2.2.5.1.1 Product-Specific Configuration

If your product supports product-specific configuration information or if it needs dynamic inks support, then your Printer Controller **should** transfer it whenever your printer determines that any product-specific configuration has changed. Since this information is product-specific, please check with your **Fiery** Product Team to know what data should be sent. To properly transfer product-specific configuration information, your Printer Controller's **Control Path** should do the following:

1. Transfer new product-specific configuration (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/oem/configuration).

2.2.5.1.2 Ink Status

If your product supports communicating information about its installed inks (and your Printer Controller's **Control Path** has already enumerated its installed inks), then your Printer Controller **should** communicate changes in ink status (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/inks/<id>/status for each ink in which a change has occurred).

2.2.5.1.3 Input Status

If your product supports communicating information about its installed inputs (and your Printer Controller's **Control Path** has already enumerated its installed inputs), then your Printer Controller **should** communicate changes in input status (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/inputs/<id>/status for each input in which a change has occurred).

2.2.5.1.4 Uniformity/Nozzle-Out Corrections

If your product supports uniformity and/or nozzle-out corrections, then your Printer Controller **should** transfer it whenever your printer determines that the DFE needs to update its compensation values. To properly transfer uniformity and/or nozzle-out corrections, your Printer Controller **should** do the following:

1. Upload the necessary nozzle-out correction files (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/nozzleOut for each set of printheads).
2. Apply uploaded nozzle-out correction files (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/nozzleOut).
3. Upload the necessary uniformity correction files (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/uniformity for each set of printheads).
4. Apply uploaded uniformity correction files (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/uniformity).

2.2.5.1.5 Calibration/Profiling Corrections

If your product has an inline sensor and supports calibration and/or profiling corrections, then your Printer Controller **should** transfer it at the end of a calibration or profiling job. To properly transfer calibration and/or profiling corrections, your Printer Controller **should** do the following:

1. Set calibration status when the printer is processing a calibration job (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/calibration/status).
2. Upload the necessary calibration correction file (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/calibration/upload).
3. Set profiling status when the printer is processing a profiling job (execute a PUT at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/profiling/status).
4. Upload the necessary profiling correction file (execute a POST at https://<fiery_ip>/fierypath/v1/system/printer/oem/corrections/profiling/upload).

2.3 Printer UI

If your product implements a **Printer UI**, it **should** support several “actions”, including the ones listed below:

- Start: used to indicate the Printer Operator wants the next available job to “start” printing.
- Stop: used to indicate the Printer Operator wants the current job to “stop” printing. Depending on the design of your Printer Controller’s **Control Path**, your Printer Controller’s **Printer UI** could support a **Stop (Immediate)** action, a **Stop (Normal)** action, or both. The **Stop (Immediate)** action and the **Stop (Normal)** action are discussed later.
- Recover: used to indicate the Printer Operator wants to “recover” any images that failed to completely print when a “stop” was requested.
- Abort: used to indicate the Printer Operator wants to “cancel” the currently printing job.

The figure below shows an example of a Printer Controller’s **Control Path** state diagram, along with the supported Printer Operator UI actions.

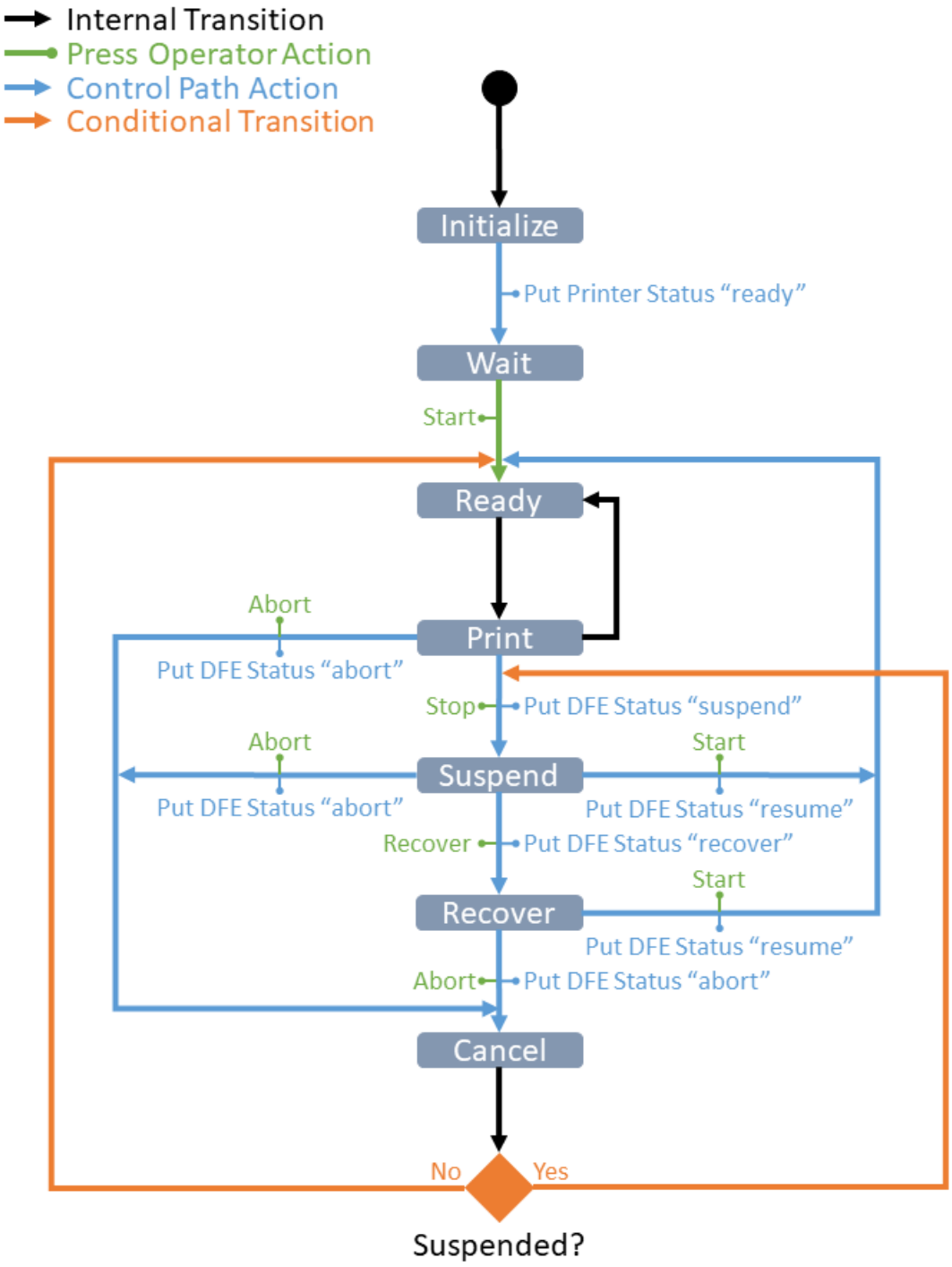


Figure 2: *Control Path state diagram and Press Operator UI actions.*

2.3.1 Start

Upon receiving the Printer Operator's **Start** action, if your Printer Controller is currently in the **Wait** state, then it will simply transition to the **Ready** state. Otherwise, the **Start** action corresponds to your Printer Controller's **Control Path** performing the **Set DFE Status** action (see the **FieryPath** interface specification) with the "action" element set to "resume". Please refer to the Suspension/Recovery sub-section under the Actions section earlier in this document for more details regarding "resume".

2.3.2 Stop

If the design of your Printer Controller supports both **Suspend (Immediate)** and **Suspend (Normal)**, then your Printer Controller's **Printer UI** should support two **Stop** actions: **Stop (Immediate)** and **Stop (Normal)**. Regardless, the **Stop** action corresponds to your Printer Controller's **Control Path** performing the **Set DFE Status** action (see the **FieryPath** interface specification) with the "action" element set to "suspend". Please refer to the Suspension/Recovery sub-section under the Actions section earlier in this document for more details regarding "suspend".

2.3.3 Recover

The **Recover** action corresponds to your Printer Controller's **Control Path** performing the **Set DFE Status** action (see the **FieryPath** interface specification) with the "action" element set to "recover". Please refer to the Suspension/Recovery sub-section under the Actions section earlier in this document for more details regarding "recover".

2.3.4 Abort

The **Abort** action corresponds to your Printer Controller's **Control Path** performing the **Set DFE Status** action (see the **FieryPath** interface specification) with the "action" element set to "abort" or "abortall". Please refer to the Suspension/Recovery sub-section under the Actions section earlier in this document for more details regarding "abort" or "abortall".

3 FAQ

The following is a list frequently asked questions (FAQ) and their answers:

Q Which language should I use to create my Printer Controller's **Control Path**?

A The **FieryPath** specification is a RESTful API. Any language that supports creating such a client can be used. This can be C/C++, Node.js, C#, Python or any such language.

Q If a RESTful API fails, should I retry?

A The answer depends on the context of the API. As an example, if "login" or "websocket" action fails, it is expected that the Fiery Path Client retry until it succeeds. However, if the response to **imageStarting** or **imageCompleting** indicates a failure, the client should check the HTTP Status Code to determine if a retry is required (a status code of 404 indicates no retry is required). A status code of 422 indicates there is an issue with one of the parameters sent, so a retry will not help until it is fixed. A status code of 500 indicates an internal server error which requires contacting your **Fiery** Product Team.

Q Is DFE guaranteed to receive requests from the printer in the order in which it sent it?

A This is more of a RESTful implementation question. Typically, a client cannot assume that the DFE will receive the requests it sends in the order in which it sends it. The client needs to check the response before sending the next request. During the printing sequence, if a given printer can send "start" responses in a different order than it receives **imageStarting**, then as stated above, they need to check the response first before sending the next one. However, if the printer will always send the "start" response in the same order, then it can use a mode in DFE (sequential printer) to not have to worry about serializing these requests. It still needs to check the response to decide if a retry is required.

Q Do I have to create a WebSocket connection?

A Yes. After a successful login, without a WebSocket connection, the Fiery Path Client will not receive any printing events until a WebSocket connection is created successfully.

Q Can I use HTTP/WS to create the client?

A No. For security reasons, we only support secure Fiery Path connection using SSL. So, a client is required to use HTTPS for RESTful API's and WSS for WebSocket connection.

Q What sequence should I follow if the Fiery Path client is restarted because of a crash or any other reason?

A The Fiery Path client would go through a normal login sequence. If the client sets its status to "initializing" as part of the login sequence, it does not have to perform any special step. The DFE will detect that the client has reconnected and it will automatically invoke the "suspend/recover/resume" sequence once the printer sets its state to "ready". Additionally, for Configurations, the client may choose to resend them. However, if the client does not set its state to "initializing", then it should invoke the "suspend/recover/resume" sequence to continue printing gracefully. It should be noted that in this case, the client may not get a **jobStarting** event again during recovery but instead get jobRecovering event and should be prepared to handle that case.

Q Should the **imageStarting** event be used to throttle images?

A No. Throttling should be handled as part of **Data Path** if at all needed. Your Printer Controller's **Control Path** should respond as soon as possible to **imageStarting** events to ensure printing at speed.

Q Is the **dfeDisconnected** event guaranteed to be sent?

A No. If DFE is restarting, it will be sent. However, if the machine hosting the DFE is rebooted, this event is not guaranteed to be sent.

Q Can I get more imageStarting events than specified by “copies” in jobStarting?

A Yes. If a job has any special pages like banner, die-cut, etc., the client will receive an **imageStarting** for each of these special images. The “copies” value in **jobStarting** reflects the number of copies in the job and does not include these special images. The client can detect this by checking for the “special” flag in the **imageStarting** event.

Q Is the “id” sent as part of **jobStarting** unique or can it change when re-printing the job?

A The job “id” is guaranteed to be unique and will not change when reprinting the job.

Q What is the difference between a suspend/resume and suspend/recover/resume sequence?

A In the suspend/resume case, upon resuming the DFE will **not** resend any previously sent **imageStarting** events. In the suspend/recover/resume case, upon resuming, the DFE will resend any previously sent **imageStarting** events for all images for which your Printer Controller’s **Control Path** had either **not** received an **imageCompleting** event or received an **imageCompleting** event, but did not respond (execute a PUT at https://<fiery_ip>/fierypath/v1/jobs/<id>/status) with a “status” JSON object containing “state” set to “complete” and “id” set to the image ID received in the corresponding **imageCompleting** event.

Q What units should I use when creating an ink for “capacity” and “level”?

A DFE doesn’t interpret these values other than using them to compute a percentage available/remaining. Therefore, your Printer Controller’s **Control Path** may choose to use any appropriate unit, but it must be the same unit for both “capacity” and “level”.

Q When should I send the response to **jobCompleting event**?

Q How do I know if all images have been transferred to the printer by the DFE?

Q Can I use the “**last**” attribute to determine if I should send the response to **jobCompleting** event?

A The general answer is that the printer should send the response to **jobCompleting** after it has received **imageCompleting** for all the images for which a “start” was issued and all images have been printed.

The implementation can be simplified to simply look at the “last” attribute sent as part of **imageCompleting** to send the response to **jobCompleting**. This of course assumes that a “complete” has already been sent for all other im**ageCompleting**.

Q. Can images part of a group have different offsets?

Q. Can images part of a group be “started” in a different order?

A. All images which are part of a group should have the same offset. Additionally, they should be “started” in the same order they were received.

Q. Once the printer initiates a “recovery”, and if there is a rush job as part of the recovery, what is the expected sequence of images?

A. Typically, the printer will receive the images in the order in which **imageStarting** was normally sent to it. However, if a rush job was initiated, then on a recovery, the printer will get the images for the rush job first before continuing with the normal job. It should be noted however that the printer can continue to get **imageCompleting** for a normal job during the time it starts getting events for a rush job.

Q. Should the oem/configuration be sent prior to any other configurations like inks, inputs etc.?

A. Yes, as a general rule, oem/configuration should be sent prior to sending any other configuration immediately after a successful login.

Q. How should the client use the dfe/info resource?

A. There are 2 parts to the dfe/info resource – version and settings. For clients that want to know the current dfe Fiery Path version, they can use the version info to maintain functionality based on version changes. Additionally, there is a printer specific optional settings JSON Object that may be passed to the client which will contain product specific settings which can be accessed by the client.

Q. What is the criteria used by DFE to report "**disconnected**", "**connected**" and "**ready**" state for "Get a List of DFE Statues"?

A. If the DFE is starting up, it will report the state as "**disconnected**". This can typically happen initially when the printer completes the "**Log-in**" action and DFE is just completing its startup phase.

Immediately there-after, the DFE will transition to the "**connected**" state. It is at this point that the printer should initiate the "**Configuration**" action. The DFE will continue to remain in the "**connected**" state until **both** the printer reports its status as "**ready**" and the DFE internally is not suspended. Once both these conditions are met, DFE will transition to the "**ready**" state. It is at this point that the printer should initiate the "**Printing**" sequence.

If at any point after that, the printer suspends the DFE, it will report its state as "**connected**". Once the printer resumes the DFE, it will report the state as "**ready**".

Q. How does the printer know the type of a job?

A. As part of "**jobStarting**", the type field indicates the type of the job. Currently, it is set to one of 3 values – "normal", "calibration", or "profiling". Printers that support inline measurement can use the type to initiate the calibration or profiling workflow to upload appropriate measurement data.

Q. How should the client use the optional flags for dfe/status?

A. There are 3 optional attributes that can be used for "Set DFE Status". All of these are applicable only when setting the "action" to either abort, abortall or recover. The "reason" attribute can be used to specify why the printer has issued an abort/abortall. The "id" attribute is only when issuing an abort and identifies the job. Finally, "responseRequired" attribute can be used so that the printer does not have to respond to the corresponding "**jobCanceling**" event. In this case, it should be set to false. If the printer does not set this attribute or sets it to true, it needs to respond to the "**jobCanceling**" event, otherwise the job will not be canceled. For usage of these options with recover, please refer to the Fiery Path specification.

Q. Why is there an update for localization? What can it be used for?

A. If the printer supports localization, it needs to create the resource by using POST. The printer should add an array of all static strings as part of this call. Any subsequent POST for localization will overwrite the previous array sent by the printer. However, at times, localization strings needs to be created dynamically. In that scenario, the update method (PUT) can be used to add new strings. Correspondingly, those updated strings can be deleted by using an option JSON object in DELETE.

Q. When does the printer know the data for an image is available?

A. The only guaranteed way for a printer to know when the data is available for an image is when it receives **imageCompleting** event for that image. The printer should not use the data being available in Data Path for one or more colorants as the cue for data being available but instead use the **imageCompleting** event and send the "complete".



Q. For back to back jobs, can the printer receive any events for a second job while the first has not completed?

A. The printer is guaranteed to get all **imageStarting** for the first job before it gets **imageStarting** for a second job (the only exception is if the second job is rush). However, as the printer starts getting **imageStarting** for a second job, the first job may still not have completed, and it can expect to receive **imageCompleting** for it during that time as well.

Q. Does **imageCompleting** guarantee that the image data has been transferred?

A. It depends on the product. In most cases, the answer is yes, however, it is best that a check be made for image data once **imageCompleting** is received. If data is not available or is corrupt, the printer should invoke the suspend/recover/resume sequence.

Q. For products that use FITS Data Path protocol, how should a disconnect in socket connection be handled?

A. When a disconnect is detected by a FITS client, it should issue a suspend, followed by a recover. At that point, it should reestablish the socket connections as per FITS protocol, flush the socket buffers and then finally issue a resume which should complete the suspend/recover/resume sequence.

Q. With Fiery Path version 3.6, there is a new “images” JSON object available as part of **jobStarting**. Is this guaranteed to be sent?

A. As noted in the specification, “images” is an optional value. It is available only when the DFE can accurately determine the various elements of the “images” object. Additionally, it does not take into account any periodic “separator” sheets – only job “separator” sheets that would be inserted “before” and/or “after” a job.

Q. What sequence should I follow if the Fiery Path client receives error 1006 with websocket connection?

A. If the Fiery Path client receives 1006 error, it should reinitiate the login sequence. As part of the login sequence, if the client sets the state to “initializing”, then recovery will be automatically initiated. If the client does not set the state to “initializing”, it can initiate a recovery on its own. Additionally, configurations should not be sent again when 1006 is received.

Q. What is the purpose of “display” attribute in response to imageCompleting event?

A. For clients that use finishing options, they may hold onto images and not respond to imageCompleting event until all images in a finishing set have printed. In this case, any application like Command WorkStation that displays progress of jobs will not have any indication that the images are being printed until the client responds. In such a situation, if the client wants applications to update the job progress, they can send 2 responses for imageCompleting event – one with the “display” attribute set to true to update job progress and then without it once the image is printed.

Q. When using the “Raw_Tiff” format for Uniformity correction data, what are the expected TIFF file dimensions and content details?

A. There are several requirements of a properly formatted TIFF file when used for Uniformity correction data. First, TIFFTAG_SAMPLESPERPIXEL (i.e. TIFF tag 277) should be set to 1 – you will have one TIFF file for each colorant that you want Uniformity correction data applied to. The TIFFTAG_BITSPERSAMPLE (i.e. TIFF tag 258) should be set based on the “precision” of your corrections data – 8 for “U8” or 16 for “U16”. The TIFFTAG_IMAGEWIDTH (i.e. TIFF tag 256) should be set to the total number of nozzles for the given colorant. The TIFFTAG_IMAGELENGTH (i.e. TIFF tag 257) should be set based on the “precision” of your corrections data – 256 for “U8” (i.e. 2^8) or 65,536 for “U16” (i.e. 2^16). Each scanline of the “image” represents the Uniformity correction data for the given pixel value (e.g. the minimum pixel value for all nozzles is represented in the 1st scanline and the maximum pixel value for all nozzles is represented in the Nth scanline).

Q. Should a response to imageCompleting event always be sent? When should the printer not send a response?

A. If an image has been printed successfully, the printer should send the “complete” response to imageCompleting. However, if there is a jam or the job is being canceled, then a response to imageCompleting should not be sent.

Q. Should the default timeout for the REST client be changed?

A. It depends. Certain REST libraries have a default timeout of 30 seconds on the client side. Some other libraries have a larger or infinite timeout. For most commands, there should be no need to adjust or set the timeout from the client side except for a couple. They are the POST commands for “inputs” and “configuration”. When the DFE is restarted and the client connects, these 2 POST commands can take a bit longer, so it is recommended that the client timeout be set to 5 minutes or more only for these commands. Once again, this is required only if the REST client library used has a lower timeout value.



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